```
    // Ensure RAG system is initialized
    const rag = await initializeRAG();

    // Perform RAG query
    const result = await rag.performRAG(input);

    const message = {
      id: generateId(),
      role: 'assistant',
      content: result.answer,
    };

    // ... rest of the implementation
  }
}
```

Calls the initialization function to get the RAG system

Performs a RAG query with the input

The `initializeRAG()` function is a critical component of our RAG implementation. It creates a singleton instance of the RAG system. When `continueConversation()` is called, it checks if the RAG system is initialized, loads the index from the specified path, and then performs a retrieval-augmented query.

DEMO

Figure 7.9 shows the web interface with a RAG-generated response, demonstrating the integration of the new functionality. The user has submitted a query about the Reims

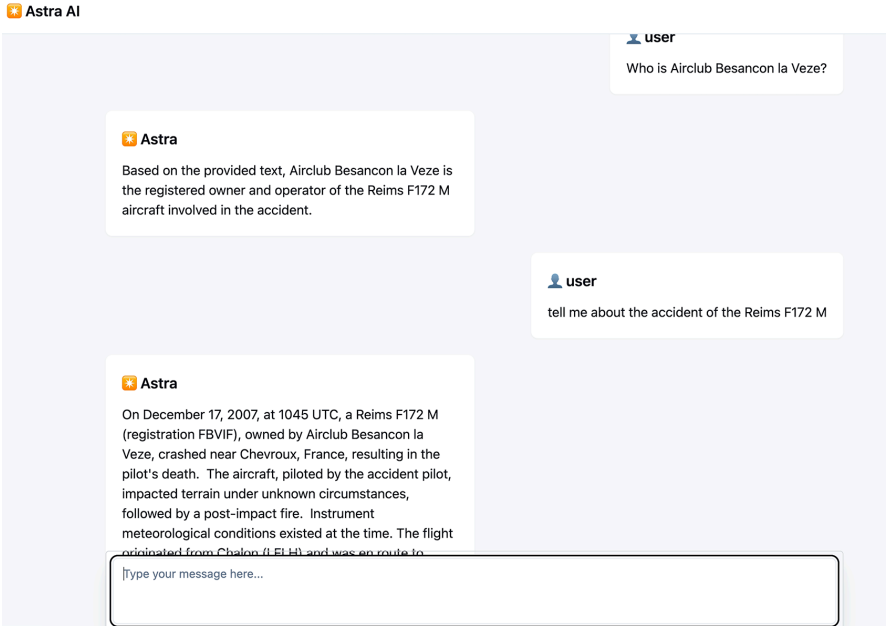


Figure 7.9 Screenshot of an Astra AI conversation interface where the user asks questions regarding the indexed list of documents using RAG